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RESEARCH ARTICLE

Time Matters: Temporal NetFlow Features for ML-Based Network Intrusion Detection

MAJED LUAY^{1,2}, SIAMAK LAYEGHY¹, NILOUFAR NOORBIN¹, MOHANAD SARHAN¹,
GAYAN KULATILLEKE¹, NOUR MOUSTAFA³, (Senior Member, IEEE),
AND MARIUS PORTMANN¹, (Member, IEEE)

¹School of Electrical Engineering and Computer Science, University of Queensland, Brisbane, QLD 4072, Australia

²College of Computing and Information Technology, University of Bisha, Bisha 61922, Saudi Arabia

³School of Engineering and Information Technology, University of New South Wales, Sydney, NSW 2052, Australia

Corresponding author: Siamak Layeghy (siamak.layeghy@uq.net.au)

ABSTRACT Most machine learning-based network intrusion detection systems (NIDS) rely on per-flow time-agnostic feature vectors, representing each flow using structural NetFlow features without explicit temporal information. This representation limits the ability to capture the intrinsic temporal structure of network traffic, such as event sequencing, inter-arrival dynamics, and non-stationary rate fluctuations. Consequently, the contribution of these temporal dependencies to detection performance remains under-characterised. In this work, we evaluate the extent to which sequential architectures, specifically Transformers, Long Short-Term Memory networks, and one-dimensional Convolutional Neural Networks, can exploit time-dependent information under controlled cross-dataset conditions. To facilitate this analysis, we introduce NF3, a temporally enriched extension of existing NetFlow-based benchmarks, which integrates fine-grained timestamps and inter-arrival-time statistics. Using these augmented datasets, we quantify temporal regularities in benign and malicious traffic and benchmark the ability of each architecture to learn time-dependent representations in both binary and multi-class settings. Our results demonstrate that augmenting conventional flow features with temporal information yields consistent gains; binary detection improves by up to 3% in F1 score, while macro-averaged multi-class F1 increases by approximately 27%, with the most significant improvements occurring in attack classes with pronounced temporal signatures. These findings indicate that incorporating time-dependent information enhances both the discriminative power and behavioural interpretability of modern NIDS.

INDEX TERMS Intrusion detection systems (IDS), NetFlow data, network traffic classification, sequential deep learning models, temporal network analysis, transformers and LSTM networks.

I. INTRODUCTION

Ensuring the security and integrity of network infrastructures has become increasingly challenging as cyber threats and large-scale attacks continue to expand in both scope and complexity [1]. For instance, in the first quarter of 2025 alone, Cloudflare reported mitigating approximately 20.5 million distributed denial of service (DDoS) attacks, in which more than 700 exceeded one terabit per second (Tbps) or one billion packets per second (Bpps) in scale [2]. The volume

and rate of traffic generated during such attacks make per-packet inspection impractical, both in terms of computational resources and storage requirements [3].

Flow-level monitoring methods, such as NetFlow [4], provide a more scalable approach by aggregating packets into summarised flows between endpoints. These flows capture various attributes of packets such as source and destination IP addresses, ports, protocols, and packet and byte counts [5]. While this reduces the volume of data to be processed, it retains the information needed for detecting anomalies and attacks [6]. Accordingly, flow-based Network Intrusion Detection Systems (NIDSs) play a

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crucial role within the broader network security ecosystem, as they enable real-time traffic monitoring and analysis to detect suspicious activities, unauthorised access attempts, and potential security breaches [7].

While traditional NIDSs primarily depend on predefined signatures to detect malicious activities [8], adversaries can often evade detection by making minor modifications to their attack patterns using publicly available tools. This highlights the need for more adaptive and intelligent detection approaches [9]. To address this limitation, modern NIDSs increasingly use machine learning (ML) techniques that learn normal network behaviour to detect anomalies, moving beyond the restrictions of static rule-based detection [1], [10], [11].

Most ML-based NIDS approaches rely on per-flow features and treat each flow as an independent instance [9], [12], which simplifies traffic representation but may overlook the temporal dimension of network activity and how events evolve over time [13], [14]. As a result, the impact of aspects such as the sequence of flows, packet inter-arrival times (IAT), sudden traffic surges, and daily fluctuations in benign activity on detection performance remains insufficiently understood [15]. Moreover, although prior studies have explored feature importance in NIDS [16], [17], [18], the specific contribution of temporal features in enhancing detection performance has received little attention.

These limitations point to the need for approaches that can capture dependencies across flows and account for how network events unfold in sequence. Recent advances in sequential learning, particularly transformer architectures, have drawn attention to the value of modelling temporal patterns in complex data [19]. In this work, we examine how such models can leverage flow sequences and temporal information to improve the detection of dynamic and sophisticated attack behaviours.

To this end, we began by augmenting existing NetFlow-based NIDS benchmark datasets [20], [21], which lack temporal information, with additional temporal features. Each flow record was enriched with start and end times, as well as IAT statistics between consecutive packets within a flow. The complete creation process for these enriched datasets, collectively referred to as **NF3**, is presented in Section III. The NF3 datasets are made publicly available to enable future research on intrusion detection methods that utilise temporal dynamics and sequential patterns [22].

Using these NF3 datasets, we performed a detailed temporal analysis of network traffic, examining how benign and various attack flows differ in their distributions of flow duration, IAT, and other timing-related properties. To assess the role of temporal features, we applied three sequential models: Long Short-Term Memory (LSTM) networks [23], one-dimensional Convolutional Neural Networks (1D-CNN) [24], and Transformers [25]. Their performance was evaluated on both binary and multi-class classification tasks, with and without temporal features. In addition,

we conducted a permutation feature importance analysis [26], which further demonstrates the value of temporal information. The main contributions of this work are as follows:

- Providing an empirical evaluation of temporal flow features for ML-based NIDS, testing their effectiveness with sequential models including LSTM, 1D-CNN, and Transformer architectures.
- Releasing the NF3 datasets,¹ an enriched version of existing NetFlow benchmarks that incorporates temporal features to enable further research on flow-based intrusion detection.
- Presenting a detailed temporal analysis of network traffic in the NF3 datasets, showing how benign and various attack classes differ in flow duration, IAT, and other timing-related attributes.

The rest of the paper is organised as follows. Section II reviews related work and identifies research gaps. Section III describes the NF3 datasets. Section IV presents the temporal analysis of network traffic. Section V reports the experimental evaluation of the effectiveness of temporal features using sequential models. Finally, Section VII concludes the paper and discusses directions for future work.

II. RELATED WORKS

Prior studies [17], [27], [28], [29] have systematically analysed feature importance in flow-based NIDS datasets. Awad et al. [17] performed correlation-based and threshold-driven selection on a large NetFlow-formatted IoT intrusion dataset containing over 17 million flows, reducing the number of features to 13 while maintaining up to 100% detection accuracy. Binbusayyis and Vaiyapuri [27] conducted a comprehensive analysis to recommend the most effective feature evaluation measure for intrusion detection. Their study compared consistency-based, correlation-based, information-theoretic (information gain), and distance-based criteria, each reflecting a different strategy for identifying informative features. The evaluation was performed on two benchmark datasets: NSL-KDD [30] and UNSW-NB15 [31], and found that the consistency-based measure provided the best performance. Their results indicated that an optimal subset of features significantly improves detection accuracy. This underscores the value of identifying impactful features, such as the temporal ones in our study. Sarhan et al. [28] applied the SHAP technique [32] to evaluate the contribution of NetFlow features and offer insight into how individual features affect model predictions. Choi et al. [29] extended this direction with an attack-specific analysis framework using NF-BoT-IoT-v2 and NF-ToN-IoT-v2 datasets, identifying dominant features for each attack type and demonstrating improved performance with reduced dimensionality.

Beyond general feature analyses at the dataset level, several works have focused on individual NetFlow features. One commonly studied feature is flow length, which provides

¹Datasets available at: https://staff.itee.uq.edu.au/marius/NIDS_datasets/

valuable information about traffic dynamics and has been the focus of extensive research [33], [34], [35]. Studies such as Iorliam et al. [33] have shown that malicious TCP flows deviate significantly in their size distribution compared to benign flows. Beigi et al. [34] employed a decision tree classifier incorporating several feature categories, including duration-based metrics, to detect botnet activity, achieving detection rates between 75% and 99%. Chen et al. [35] analysed slow-paced and persistent network attacks and showed that malicious flows exhibit distinct duration-based and length-distribution patterns that deviate from those of benign traffic.

IAT features have also received considerable attention due to their ability to capture fine-grained temporal patterns [36], [37], [38], [39], [40]. Benson et al. [38] analysed IAT across ten diverse data centre networks spanning universities, enterprises and cloud providers, revealing variations tied to different application-layer behaviours. Building on this, Kandula et al. [39] examined the distribution of key traffic features using a multi-level monitoring approach, combining SNMP counters, sampled flow or packet headers, and deep packet inspection to capture a comprehensive view of traffic volume, congestion, and application-level dynamics.

Nonetheless, the temporal analysis of standard flow formats, like NetFlow [4], remains under-explored [41]. Studies have explored the effectiveness of sequential learning models, such as LSTM, in extracting temporal characteristics from NetFlow data for NIDS [14]. Some researchers adopted CNN and LSTM models simultaneously to construct a hybrid model [42], [43]. CNNs are widely used for spatial feature extraction, particularly in computer vision, and have been adapted for traffic modelling [44]. In [42] and [43], the authors introduce the Spatial and Temporal Aware Intrusion Detection Model (STIDM). STIDM is a spatio-temporal feature extraction model designed to analyse IAT features between consecutive packets. This model employs a well-known CNN architecture, LeNet-5, to extract spatial features, complemented by a modified LSTM to capture temporal patterns. Although this method allows grouping of packets into flows, it does not effectively facilitate the determination of broader temporal patterns across NetFlow data.

Corsini et al. [14] explored temporal sequences of network traffic flows that denote patterns of malicious activities. The main focus was to identify whether specific temporal patterns exist for each attack class, rather than competing with state-of-the-art solutions. The paper investigates the use of LSTM to learn temporal patterns in network flows and compares the performance of the LSTM to a static Feed-forward Neural Network (FNN) model.

These ad-hoc methods lead to inconsistent feature definitions, making it difficult to reproduce results or compare models reliably. Furthermore, researchers who generate or augment timestamps in their datasets (e.g., [45], [46]) often restrict the use of these modified datasets to their own experiments, without making them publicly available.

Furthermore, no publicly available NetFlow-formatted NIDS benchmark currently includes temporal features, which has prevented a systematic investigation of their impact on detection performance. This critical gap forms the primary motivation for our work: to rigorously evaluate the role and importance of temporal information in flow-based intrusion detection. To address this gap, we introduce NF3; the third generation of the standardised NetFlow benchmark datasets, expanded with explicit temporal features useful for exploring time-aware and sequence-based intrusion detection methods.

III. NF3: NetFlow DATASETS WITH TEMPORAL FEATURES

This section outlines the structure and preparation pipeline of the NF3 datasets. Building on the methodology used to generate the earlier versions [20], [21], NF3 extends the datasets by introducing a temporal dimension through the inclusion of fine-grained timestamps and IAT features.

A. NEW FEATURES

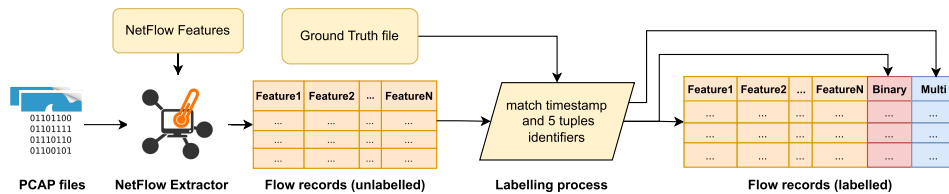
Table 1 summarises the features included in this version of the datasets alongside those of previous versions. As shown, the new release retains all the features of the previous versions and introduces an additional set of ten temporal features. These new features fall into two categories: timestamp features (highlighted in green), which specify the start and end times of flows in milliseconds, and IAT features (highlighted in blue), which capture various statistical measures of the inter-arrival times between consecutive packets within a flow.

Timestamps allow for the precise ordering of events within a network. They are also helpful in accurately calculating flow durations, which are important for detecting abnormally long or short connections that might indicate malicious activity [47]. Historical timestamp data allows for the creation of baselines of normal network behaviour (e.g., typical traffic patterns during different times of day [38], [40]). Deviations from these baselines can then be identified as anomalies. This temporal granularity is necessary for the advanced analyses and correlations that will be explored in the subsequent sections.

IAT is a key temporal metric that measures the elapsed time between consecutive packets at a monitoring point [40]. When calculated in a specific direction, either from source to destination or in reverse, IAT exposes fine-grained patterns in traffic behaviour, such as flow regularity, congestion, and burstiness [36]. Although IAT is defined at the packet level, it becomes particularly informative when aggregated at the flow level. To compute this metric accurately, each packet's timestamp is recorded upon arrival, and the difference between consecutive timestamps is determined. These time differences are then aggregated to obtain the minimum, maximum, average, and standard deviation for each flow. The resulting set of metrics forms a concise temporal signature that can uncover traffic anomalies not evident from volume or structural features alone [37].

TABLE 1. Feature availability across NF1, NF2, and NF3 versions. Newly added temporal features are highlighted: timestamps (Green), and statistics for IATs (Blue).

Feature Name	Description	NF1 [20]	NF2 [21]	NF3
IPV4_SRC_ADDR	IPv4 source address	✓	✓	✓
IPV4_DST_ADDR	IPv4 destination address	✓	✓	✓
L4_SRC_PORT	Source port number	✓	✓	✓
L4_DST_PORT	Destination port number	✓	✓	✓
PROTOCOL	IP protocol identifier	✓	✓	✓
L7_PROTO	Application protocol (numeric)	✓	✓	✓
IN_BYTES	Incoming number of bytes	✓	✓	✓
OUT_BYTES	Outgoing number of bytes	✓	✓	✓
IN_PKTS	Incoming number of packets	✓	✓	✓
OUT_PKTS	Outgoing number of packets	✓	✓	✓
TCP_FLAGS	Cumulative TCP flags	-	✓	✓
CLIENT_TCP_FLAGS	TCP flags set by client	-	✓	✓
SERVER_TCP_FLAGS	TCP flags set by server	-	✓	✓
MIN_TTL	Minimum TTL in flow	-	✓	✓
MAX_TTL	Maximum TTL in flow	-	✓	✓
LONGEST_FLOW_PKT	Longest packet in flow (bytes)	-	✓	✓
SHORTEST_FLOW_PKT	Shortest packet in flow (bytes)	-	✓	✓
MIN_IP_PKT_LEN	Minimum IP packet length	-	✓	✓
MAX_IP_PKT_LEN	Maximum IP packet length	-	✓	✓
SRC_TO_DST_SECOND_BYTES	Bytes/sec (src→dst)	-	✓	✓
DST_TO_SRC_SECOND_BYTES	Bytes/sec (dst→src)	-	✓	✓
RETRANSMITTED_IN_BYTES	Retransmitted bytes (src→dst)	-	✓	✓
RETRANSMITTED_IN_PKTS	Retransmitted packets (src→dst)	-	✓	✓
RETRANSMITTED_OUT_BYTES	Retransmitted bytes (dst→src)	-	✓	✓
RETRANSMITTED_OUT_PKTS	Retransmitted packets (dst→src)	-	✓	✓
SRC_TO_DST_AVG_THROUGHPUT	Avg. throughput (src→dst)	-	✓	✓
DST_TO_SRC_AVG_THROUGHPUT	Avg. throughput (dst→src)	-	✓	✓
NUM_PKTS_UP_TO_128_BYTES	IP size ≤ 128	-	✓	✓
NUM_PKTS_128_TO_256_BYTES	IP size 128–256	-	✓	✓
NUM_PKTS_256_TO_512_BYTES	IP size 256–512	-	✓	✓
NUM_PKTS_512_TO_1024_BYTES	IP size 512–1024	-	✓	✓
NUM_PKTS_1024_TO_1514_BYTES	IP size 1024–1514	-	✓	✓
TCP_WIN_MAX_IN	Max TCP window (src→dst)	-	✓	✓
TCP_WIN_MAX_OUT	Max TCP window (dst→src)	-	✓	✓
ICMP_TYPE	ICMP type × 256 + ICMP code	-	✓	✓
ICMP_IPV4_TYPE	ICMP Type only	-	✓	✓
DNS_QUERY_ID	DNS transaction ID	-	✓	✓
DNS_QUERY_TYPE	DNS query type	-	✓	✓
DNS_TTL_ANSWER	TTL of DNS A record	-	✓	✓
FTP_COMMAND_RET_CODE	FTP client return code	-	✓	✓
FLOW_DURATION_MILLISECONDS	Flow duration (milliseconds)	-	✓	✓
DURATION_IN	Duration (client→server)	-	✓	✓
DURATION_OUT	Duration (server→client)	-	✓	✓
FLOW_START_MILLISECONDS	Flow start time (milliseconds)	-	-	✓
FLOW_END_MILLISECONDS	Flow end time (milliseconds)	-	-	✓
SRC_TO_DST_IAT_MIN	Min IAT (src→dst)	-	-	✓
SRC_TO_DST_IAT_MAX	Max IAT (src→dst)	-	-	✓
SRC_TO_DST_IAT_AVG	Average IAT (src→dst)	-	-	✓
SRC_TO_DST_IAT_STDDEV	Standard deviation IAT (src→dst)	-	-	✓
DST_TO_SRC_IAT_MIN	Min IAT (dst→src)	-	-	✓
DST_TO_SRC_IAT_MAX	Max IAT (dst→src)	-	-	✓
DST_TO_SRC_IAT_AVG	Average IAT (dst→src)	-	-	✓
DST_TO_SRC_IAT_STDDEV	Standard deviation IAT (dst→src)	-	-	✓

**FIGURE 1.** Workflow used to prepare the NF3 datasets. Raw pcap files are first converted into flow-level records using the nProbe tool. These unlabelled flows are then matched with external ground truth files containing attack events annotated with 5-tuples and precise timestamps. A matching step aligns flows to attacks based on protocol, source/destination IPs and ports, and temporal overlap. The final labelled NetFlow dataset includes NetFlow features and both binary and multi-class labels. In this work, we follow the NF2 [21] for dataset creation, and incorporate IAT and flow statistics to provide complementary information for complex flow understanding.

B. DATASET CREATION

The methodology used to generate NF3 datasets follows the same core workflow as in previous releases [20], [21]. Raw traffic captures from the source datasets [31], [48],

[49], [50] were converted into NetFlow records using the nProbe tool [51]. nProbe operates as a NetFlow/IPFIX exporter that aggregates packets into flow-level summaries. The exporter was configured with an active timeout of two

TABLE 2. Statistics of attack types across the datasets, showing the count of flows categorised under each attack and benign class.

Attack Type	NF-UNSW-NB15-v3	NF-CSE-CIC-IDS2018-v3	NF-ToN-IoT-v3	NF-BoT-IoT-v3
Benign	2,237,731	17,514,626	16,792,214	51,989
DoS	5,980	302,966	203,456	8,034,190
DDoS	—	1,324,350	4,141,256	7,150,882
Reconnaissance	17,074	—	—	1,695,132
Backdoor	4,659	—	203,384	—
Fuzzers	33,816	—	—	—
Exploits	42,748	—	—	—
Analysis	1,226	—	—	—
Generic	19,651	—	—	—
Shellcode	2,381	—	—	—
Worms	158	—	—	—
Web Attacks	—	2,538	—	—
Infiltration	—	188,152	—	—
BoT	—	207,703	—	—
BruteForce	—	575,194	—	—
Scanning	—	—	1,358,977	—
XSS	—	—	2,834,435	—
Password	—	—	1,594,777	—
Injection	—	—	381,777	—
Ransomware	—	—	3,971	—
MITM	—	—	6,013	—
Theft	—	—	—	1,615
Total	2,850,806	20,115,529	27,520,260	16,881,819

minutes and the default inactive timeout to segment long-lived flows while preserving temporal resolution. The option `-dont-reforge-timestamps` was enabled to maintain the original packet timestamps rather than the exporter's emission times. This configuration ensured that the generated flow records accurately reflected the timing behaviour of the captured traffic and supported precise temporal analysis. Flow records were aligned with ground-truth annotations provided by the original datasets using protocol, source and destination IP addresses, ports, and timestamp overlap. Each matching step produced binary and multi-class labels that identify benign traffic and specific attack categories. For the binary label, each record is assigned a value of 0 for benign and 1 for malicious flows. The ratio of benign to malicious traffic varies notably across the NF3 datasets: NF-UNSW-NB15-v3 includes about 5.4 % malicious flows, NF-CSE-CIC-IDS2018-v3 about 12.9 %, NF-ToN-IoT-v3 around 39 %, and NF-BoT-IoT-v3 is dominated by attack traffic with nearly 99.7 % malicious flows. The detailed distribution of individual attack classes across the NF3 datasets is presented in Table 2. All processed datasets were unified under a common feature schema to ensure consistency, comparability, and reproducibility across NF3 releases. The overall conversion and labelling workflow is illustrated in Figure 1.

IV. TEMPORAL ANALYSIS OF NIDS BENCHMARK DATASETS

In this section, we conduct an in-depth, dataset-level temporal analysis of patterns that are not visible through structural or volume-based features alone. Our focus here is exploratory to highlight how attributes such as flow durations and IAT vary across attack types and datasets. These descriptive insights help us understand the datasets better before applying any evaluation models. To simplify notation in the subsequent analysis and model evaluation, the NF3 datasets are referred to using the abbreviated names: NF-UNSW, NF-CIC,

NF-ToN, and NF-BoT corresponding to NF-UNSW-NB15-v3, NF-CSE-CIC-IDS2018-v3, NF-ToN-IoT-v3, and NF-BoT-IoT-v3, respectively.

A. FLOW DURATION DISTRIBUTION (FDD)

Flow duration is a fundamental temporal attribute that represents how long a communication persists between endpoints [47]. Assessing these distributions helps to reveal behavioural differences between benign and malicious traffic, such as short bursts in scans versus prolonged sessions in data exfiltration. Here, we assess the flow duration distributions to investigate how different attack classes vary in the length of their network sessions and whether these temporal patterns diverge from benign behaviour. As shown in Figure 2, each histogram illustrates the frequency of flows across logarithmically scaled duration ranges for all traffic classes in the four NF3 datasets. Flow durations are capped at two-minute intervals to align with the NetFlow exporter configuration.

In NF-UNSW, the overall FDD is dominated by short-lived flows across nearly all classes. Certain attack classes, such as Worms and Analysis, remain strictly confined below two or three seconds. In NF-CIC, the distribution becomes more diverse. The benign traffic shows a uniform spread across the entire duration range. DoS and Infiltration attacks also extend across nearly the entire durations. Other attack types are mostly concentrated below one minute. In NF-ToN, although it includes nine attack classes, the majority of flows occur within very short durations. Only Password and MITM attacks, along with benign traffic, extend into longer duration ranges. The remaining attacks are tightly clustered near the lower duration bins. Finally, in NF-BoT, there is no clear distinction among attack classes in terms of flow duration. The distribution shows that certain duration bins contain a dominant number of flows, while comparatively fewer flows appear in other ranges.

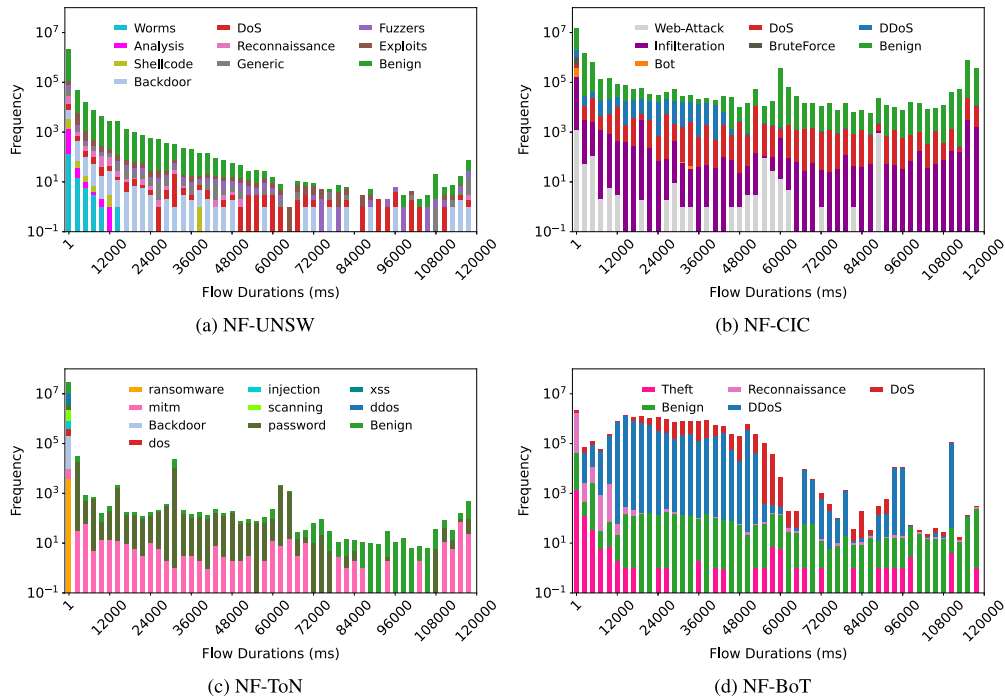


FIGURE 2. FDD across NF3 datasets. Each subplot visualises the distribution of flow durations. Bars are stacked by class and larger class counts appear behind smaller ones.

From these observations, several conclusions can be drawn. The benign traffic consistently exhibits a uniform distribution across the full duration range, which reflects diverse and naturally varying communication patterns. In contrast, DoS and DDoS attacks show variable duration behaviour, which may arise from differences in attack tools, configurations, and rate parameters that manipulate how long each flow persists. Scanning attacks are characteristically very short, aligning with their purpose of rapidly probing multiple targets without maintaining sustained connections. Minor inconsistencies observed across datasets could also stem from their synthetic or replayed nature, which may influence how durations are recorded or segmented during capture. Overall, flow duration emerges as a potentially discriminative temporal feature for some attack classes, which can be a valuable cue for understanding their persistence and behavioural dynamics.

B. ANALYSIS OF PACKET INTER-ARRIVAL TIME

We evaluate IAT features to examine how packet-level timing differs between benign and attack traffic across the NF3 datasets. We focus on the average IAT in both source-to-destination and destination-to-source directions to capture the overall transmission rhythm of each class. Flows with zero average IAT values were removed from the assessment to avoid bias and noise in the distribution.

Figure 3 shows the distribution of average IATs for all four NF3 datasets. NF-UNSW shows a clear separation between benign and attack flows, which makes it a relatively easy binary classification case. The benign traffic spans a

wider range of IAT values, whereas most attack classes exhibit a cluster within narrow, low-IAT regions, indicating stable and repetitive packet timing. In contrast, NF-CIC exhibits a more varied temporal behaviour. The DDoS flows generally have shorter IAT than benign traffic, particularly in the destination-to-source direction. Bot and Brute Force attacks show a very clear separation from normal traffic, with values tightly clustered within a narrow IAT range. Interestingly, most DoS flows are clustered within average IAT exceeding one minute with narrow timing intervals despite the overlap with benign traffic.

In NF-ToN, the average IAT distribution allows many attack types to be easily distinguished from one another. Several classes form distinct, non-overlapping clusters that reflect clear temporal differences between their traffic patterns. The Backdoor flows exhibit tightly clustered IAT values. Other attack types, such as DoS, DDoS, and Scanning, also concentrate near the lower IAT ranges. A similar pattern appears in NF-BoT, where the DoS and DDoS classes demonstrate a significant separation from benign and other attack types, despite their large share of total flows. This distinction highlights the strong temporal regularity of flooding-based attacks in the dataset, making the average IAT a particularly discriminative feature to identify them.

Across all NF3 datasets, average IAT proves to be a strong temporal discriminator. Flooding attacks can be identified by their dense concentration of extremely low IATs, whereas persistent or stealthy attacks occupy longer but still structured timing ranges. Benign traffic, in contrast, is characterised by irregular and widely dispersed IATs that lack repetitive

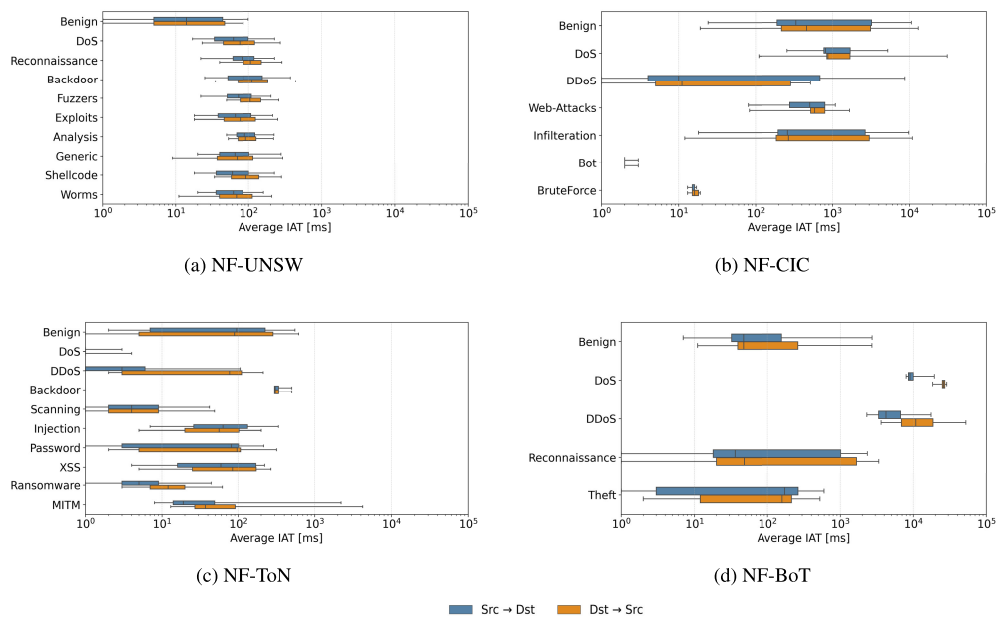


FIGURE 3. Distribution of average IAT across different attack classes in NF3 datasets. The x-axis represents the average IAT while the y-axis corresponds to the attack categories.

temporal patterns. This consistent difference between the temporal regularity of attacks and the variability of benign flows demonstrates that average IAT serves as a reliable and generalisable feature for distinguishing automated or tool-driven activity from normal network behaviour.

Although this analysis focuses on the average IAT, the other statistics introduced in the NF3 datasets—minimum, maximum, and standard deviation—also hold strong potential for intrusion detection. The minimum IAT can capture micro-burst behaviours where packets are transmitted in very rapid succession, which is typical in high-rate flooding or scanning attacks. The maximum IAT reflects the longest idle periods within a flow and may indicate session persistence or delayed responses often seen in backdoor or Infiltration activities. The standard deviation of IAT measures the stability of packet timing and can therefore distinguish between steady automated traffic and irregular human-driven communication. These complementary statistics provide additional dimensions for modelling temporal behaviour and can enhance the discriminative power of time-aware intrusion detection systems when combined with the average IAT feature.

C. TEMPORAL DISTRIBUTION OF FLOWS

When analysing traffic over time, it is important to track how attack classes are distributed across different time intervals. Identifying the exact occurrence periods of each attack class helps validate the correctness of ground-truth labels and exposes how attacks are scheduled within the datasets. This information clarifies whether attacks appear as isolated events or overlap with benign activity, which directly influences the difficulty of learning and detection. In this subsection, we represent the flow rates as a time

series for the flows of each attack class to identify their exact occurrence periods. Across the NF3 datasets, most attack classes were executed independently and can be observed as stand-alone events within benign background traffic. However, NF-UNSW is an exception, where all attacks are launched simultaneously. While such overlap can occur in real-world scenarios, analysing each attack class individually is often more informative for capturing distinct behaviours. Figure 4 illustrates one representative day from each NF3 dataset, where flows are aggregated per minute and plotted on a logarithmic scale to improve visual clarity.

Using this representation, volumetric attacks such as DoS and DDoS are clearly visible in some datasets, particularly in NF-CIC and NF-BoT, where the malicious flow rate rises sharply above benign activity, forming distinct temporal spikes. In contrast, similar attack classes in other datasets appear less pronounced, likely due to slower or low-intensity variants of DoS behaviour that blend more smoothly with background traffic. These observations emphasise that, beyond identifying when attacks occur, analysing the temporal intensity of flow activity is crucial for understanding how different attack types manifest in network behaviour.

Building on this observation, it is also valuable to explore how other feature values evolve temporally within each traffic class. Features such as byte and packet volumes can reveal deeper insights into the behaviour and intensity of network activities [52]. This temporal perspective helps identify fluctuations that may correspond to distinct attack stages or operational patterns.

Figure 5 illustrates this by showing the IN_BYTES feature aggregated per minute for the same representative day used earlier. Each time series corresponds to a specific attack class, enabling visual comparison of their activity levels

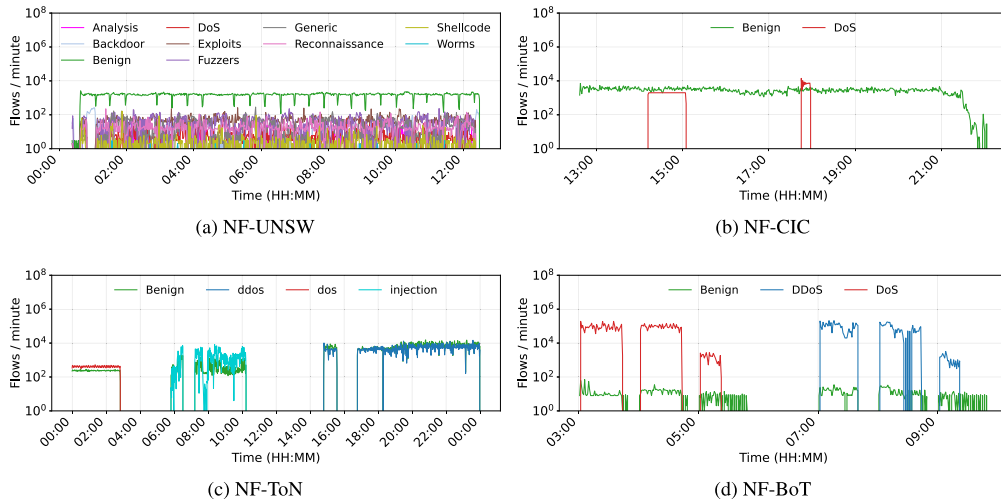


FIGURE 4. Temporal distribution of traffic flow rates across NF3 datasets. The x-axis represents the hours of the day, and the y-axis indicates the total number of flows per minute for each attack class.

and temporal intensity. Monitoring such numerical features over time helps uncover distinctive traffic behaviours and can reveal periods of abnormal data transfer associated with different attack patterns.

A similar observation to that in the flow-rate emerges here. Not all volume-based attacks produce consistently high byte counts; some DoS flows show low or moderate volumes that overlap with benign traffic. This again indicates that attack traffic can be intentionally modulated to resemble normal activity, as in slow-rate DoS variants that sustain long-lived but low-throughput connections. Such recurring patterns reinforce that effective detection requires attention to temporal persistence and behavioural dynamics, not merely the instantaneous flow volume.

V. EVALUATION PIPELINE

To investigate the role of the temporal features in attack detection, we evaluate their effectiveness using three sequential deep learning models, namely LSTM, 1D-CNN, and Transformer architectures.

Preprocessing: The evaluation begins with a structured preprocessing step to prepare the NetFlow data for sequential learning. Separate procedures are used for numerical and categorical attributes.

We apply MinMax normalisation to scale all numeric NetFlow features to the $[0, 1]$ range. Categorical fields are processed using One-Hot Encoding (OHE) for the top 20 most frequent values. Specifically, the top 20 are identified and encoded as separate binary values, while all other values are grouped into a single category labelled *Other*. This method allows the model to capture patterns from common categorical values while maintaining computational efficiency. The encoding strategy follows best practices established in prior work that investigated preprocessing strategies for categorical data in NIDS settings [53].

The timestamp features `FLOW_START_MILLISECONDS` and `FLOW_END_MILLISECONDS` were originally recorded

in Unix time (i.e., milliseconds since 1970), resulting in large absolute values. These are not meaningful for flow-level learning and can negatively impact model stability. Therefore, we first shift these values so that the earliest flow starts at 0 milliseconds, aligning all flows relative to the beginning of the dataset. The shifted values are then scaled to $[0, 1]$ using MinMax normalisation.

To prevent biased learning and ensure a fair, generalisable evaluation, we exclude non-temporal features that are known to correlate strongly with specific attack types [17], [54]. Previous studies [21], [54] have shown that including identifiers such as IP addresses and port numbers can artificially boost model performance by allowing models to learn from dataset-specific artefacts rather than genuine behavioural patterns. In particular, [21], [54] reported biased prediction outcomes when Time-to-Live (TTL) features were included in the UNSW-NB15 dataset. Although this issue was specific to the UNSW-NB15 dataset, we removed TTL features from all four NF3 datasets to maintain consistency across evaluations. Accordingly, we exclude all features related to IP addresses, port numbers, and TTL values to promote generalisation and ensure the models focus on broader traffic behaviours rather than memorising flow-specific details.

Sequence Construction: The datasets are transformed into fixed-length sequences before being fed into the models. Each sequence consists of N consecutive flows, preserving the natural order of NetFlow records as they appear in the datasets. This ordering follows the chronological sequence of flow observations and enables the models to capture temporal dependencies across multiple flows. Rather than assigning a single label to an entire sequence, this study adopts a sequence labelling strategy, where each element (i.e. each flow) retains its original class label within the sequence [55]. Essentially, the model then takes into consideration the temporally adjacent neighbour flows.

To determine an appropriate sequence length N , an exploratory study was conducted by evaluating model

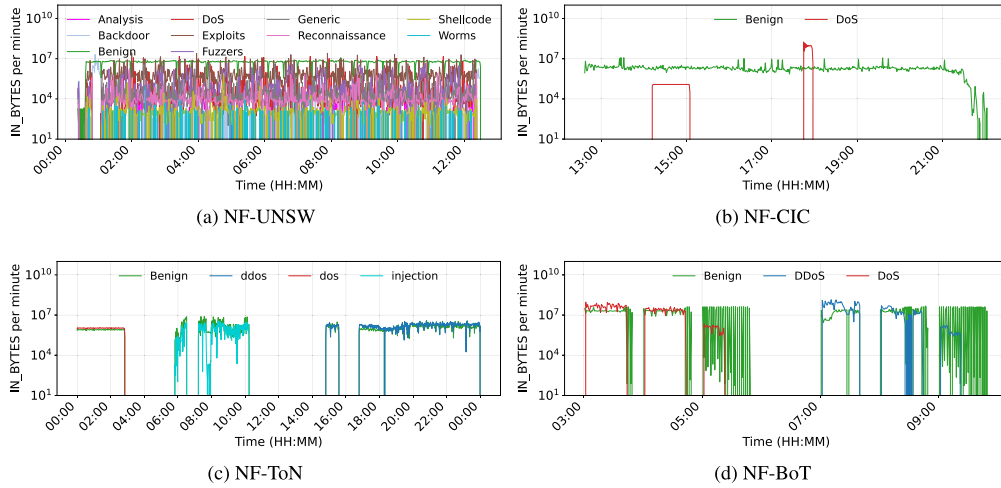


FIGURE 5. Time series of the IN_BYTES feature aggregated per minute for one representative day from each NF3 dataset. The x-axis represents the hours of the day, and the y-axis indicates the total incoming bytes per minute for each attack class.

performance across different sequence lengths from 10 to 100. A sequence length of $N = 10$ yielded the highest F1 scores, while performance gradually decreased as the sequence length increased. Longer sequences span a wider range of traffic behaviours within a single sample, which can dilute localised temporal patterns relevant for detection. Sequence lengths below 10 flows were not examined, as they provide limited temporal context. Based on these observations, $N = 10$ was selected for the experiments.

Model Architectures: The generated flow windows form the input sequences for three sequential models: LSTM, 1D-CNN, and Transformer. All models use the Adam optimiser (learning rate = 0.0001) and a time-distributed dense output layer with sigmoid or softmax activation depending on the task. Models were trained for 10 epochs with a batch size of 16, using early stopping based on validation loss (patience = 3) and restoring the best-performing weights. The time-distributed layer enables the model to produce a prediction for each flow within the sequence. The LSTM employs four bidirectional recurrent layers of ten units each to capture sequential correlations. The 1D-CNN consists of three convolutional layers with 64, 128, and 256 filters (kernel size = 3, ReLU activation, batch normalisation) to extract local temporal patterns. The Transformer includes four encoder layers (hidden dimension = 64, two attention heads, ReLU feed-forward blocks, dropout, and residual connections) with learned positional encoding to model long-range dependencies. This unified configuration isolates the influence of temporal features while keeping the architectural complexity comparable.

Finally, the trained models are subsequently evaluated using a shuffling-based permutation analysis to investigate the role of temporal features in influencing model decisions.

VI. RESULTS

To evaluate the impact of newly introduced features, two complementary evaluations are performed: (i) comparing the

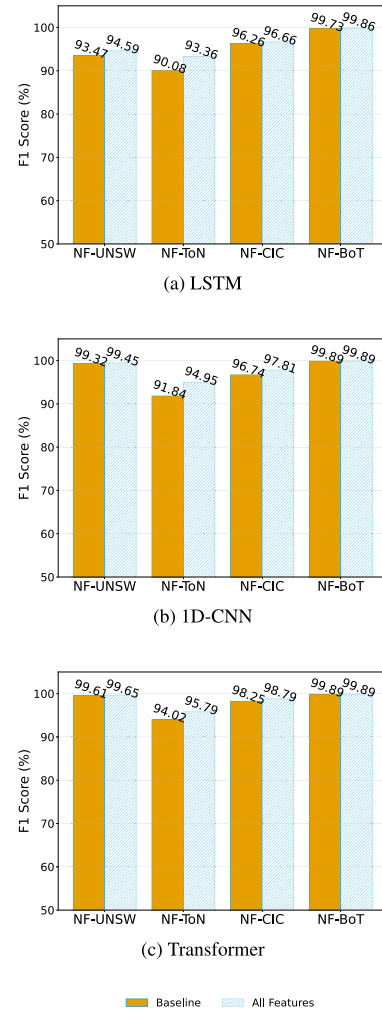


FIGURE 6. Results of binary classification.

performance of binary and multi-class classification tasks when excluding and including them, and (ii) permutation

TABLE 3. Per-class F1 scores (%) for NF-UNSW with and without temporal features.

Class	LSTM		1D-CNN		Transformer	
	With	Without	With	Without	With	Without
Benign	36.84	73.81	98.84	99.26	99.70	99.78
Analysis	0.19	0.30	31.32	31.33	34.23	36.62
Backdoor	0.00	1.09	8.32	10.55	13.10	17.25
DoS	0.72	0.99	12.12	13.18	13.79	17.25
Exploits	3.41	9.91	41.87	39.72	51.13	49.41
Fuzzers	2.71	2.28	42.80	39.26	38.92	39.30
Generic	3.27	4.87	52.71	57.39	66.17	72.46
Reconnaissance	2.46	1.23	34.61	31.72	50.45	51.51
Shellcode	0.35	0.53	9.03	8.72	16.79	14.94
Worms	0.01	0.03	1.46	2.28	0.91	0.96
Macro Average	5.00	9.50	33.31	33.34	38.52	39.95

Note: Performance on some classes may be affected by limited sample sizes

TABLE 4. Per-class F1 scores (%) for NF-ToN with and without temporal features.

Class	LSTM		1D-CNN		Transformer	
	With	Without	With	Without	With	Without
Backdoor	68.95	66.25	99.11	99.38	99.09	99.18
Benign	69.08	79.39	83.09	92.09	85.69	92.36
DDoS	80.38	85.20	95.12	98.24	95.76	98.51
DoS	34.75	78.16	76.83	98.21	77.49	98.48
Injection	58.79	89.31	82.30	96.91	89.44	96.15
MITM	4.14	19.04	14.61	46.09	15.30	53.70
Password	88.80	95.21	94.69	96.26	94.52	96.85
Ransomware	2.87	11.64	11.43	34.54	13.13	27.71
Scanning	32.27	43.76	37.28	57.21	41.07	56.80
XSS	84.90	94.41	93.96	96.76	94.36	96.97
Macro Average	52.49	66.24	68.84	81.57	70.59	81.67

Note: Performance on some classes may be affected by limited sample sizes

analysis to assess the reliance of models on these features. In both cases, we focus primarily on macro F1 score as our evaluation metric, due to the imbalanced nature of the datasets.

A. BINARY CLASSIFICATION RESULTS

Figure 6 shows the F1 scores for the three models across all datasets. In each figure, the orange and blue bars show the F1 score without (baseline) and with temporal features, respectively. For LSTM, incorporating temporal features improves performance across all datasets. The gains are particularly notable in NF-ToN (+3.28%) and NF-UNSW (+1.12%), while more modest improvements are observed for NF-CIC and NF-BoT.

The 1D-CNN also shows consistent improvements when temporal features are added. Again, the largest gains appear in NF-ToN (+3.11%) and NF-CIC (+1.07%), whereas the NF-BoT dataset remains unchanged. Similarly, the Transformer architecture benefits from temporal features, achieving the strongest improvements in NF-ToN (+2.20%) and NF-CIC (+0.94%). The performance on NF-BoT remains saturated at 99.89%.

To further examine the impact of temporal features, we extend our analysis to the multi-class setting, where different attack categories are considered separately. This enables us to assess which attack types, if any, benefit more from these temporal features.

B. MULTI-CLASS CLASSIFICATION

For the sake of clarity and the fact that the attack classes are different in each dataset, we have presented the results related to each dataset separately. Table 3 shows the results of multi-class classification for the NF-UNSW dataset. As can be seen, while the inclusion of temporal features boosts classification performance for the Benign class in the LSTM model (from 36.84% to 73.81%), it has a marginal impact on the rest of the classes except Backdoor, DoS, and Generic, which benefit from these features across all models, with the Transformer achieving consistent gains (e.g., Backdoor: 13.10% → 17.25%; Generic: 66.17% → 72.46%). For the Transformer architecture and 1D-CNN model, while the performance for the Benign class is already very good in the baseline, i.e. hard to improve further, the rest of the classes follow a similar pattern.

Nevertheless, the overall classification performance on NF-UNSW remains poor, particularly for minority attack classes. For instance, the Worms and Shellcode classes achieve near-zero detection performance due to the extremely small number of flows of these classes (e.g., Worms, 158 flows), limiting the models' capacity to capture attack-specific patterns. Baseline multi-class classification results for this dataset remain poor, in line with prior studies [54].

Table 4 shows the results of multi-class classification on the NF-ToN dataset. As can be seen, the addition of temporal features results in substantial improvement in detecting

TABLE 5. Per-class F1 scores (%) for NF-CIC with and without temporal features.

Class	LSTM		1D-CNN		Transformer	
	With	Without	With	Without	With	Without
Benign	67.38	72.93	92.98	95.17	87.17	92.52
Bot	20.37	30.55	96.10	98.66	99.54	99.30
BruteForce	43.39	72.08	56.45	99.87	80.91	99.30
DDoS	88.16	89.19	97.87	99.06	99.09	98.16
DoS	25.70	51.30	58.51	97.89	72.35	98.19
Infiltration	4.52	8.71	12.66	18.87	6.97	13.83
Web-Attack	0.16	0.35	2.73	8.48	9.06	6.75
Macro Average	35.67	46.44	59.62	74.00	65.01	72.67

Note: Performance on some classes may be affected by limited sample sizes

TABLE 6. Per-class F1 scores (%) for NF-BoT with and without temporal features.

Class	LSTM		1D-CNN		Transformer	
	With	Without	With	Without	With	Without
Benign	26.49	20.38	25.92	34.47	31.09	32.15
DDoS	53.56	87.76	74.84	99.80	57.59	99.21
DoS	59.70	89.26	77.07	99.83	72.37	99.29
Reconnaissance	86.91	90.86	94.11	94.02	92.39	93.47
Theft	2.43	76.29	12.21	9.12	4.29	82.82
Macro Average	45.82	72.91	56.83	67.45	68.00	81.39

Note: Performance on some classes may be affected by limited sample sizes

several attack types, particularly those that were challenging under baseline features. For instance, in the LSTM model, DoS increases from 34.75% to 78.16%. The 1D-CNN model also shows improvement, especially for DoS (76.83% → 98.21%) and Injection (82.30% → 96.91%). Similar improvements can be seen in the Transformer architecture, where DoS reaches 98.48% from 77.49% and MITM 53.70% from 15.30%, demonstrating that temporal features help capture nuanced temporal patterns critical for detecting these attacks.

Table 5 shows the results of multi-class classification for the NF-CIC dataset. Similar to previous datasets, temporal information consistently enhances detection performance for multiple attack classes, including some of those with low baseline scores. In the case of the LSTM model, BruteForce detection improves from 43.39% to 72.08%, DoS from 25.70% to 51.30%, Bot from 20.37% to 30.55%, and Infiltration from 4.52% to 8.71%. The 1D-CNN model benefits even more strongly, with BruteForce reaching 99.87% from 56.45%, DoS from 58.51% to 97.89%, and Web-Attack from 2.73% to 8.48%, a clear indication of the impact of the temporal features in the detection of such attacks. Finally, the Transformer architecture also shows a similar trend, with the exception of the Web-Attack class that shows a decline in detection performance compared to the baseline, i.e. when temporal features are not included in the analysis.

The NF-BoT dataset is the last in this analysis. It has an unusual imbalance towards malicious flows where the Benign traffic constitutes less than 1% of the total data. Despite this, temporal features still lead to substantial improvements for several attack classes, as shown in Table 6. Here, the Transformer architecture shows the greatest performance improvement compared to the other two models. It provides consistent gains across all classes, with particularly large improvements for DDoS and Theft, reaching 41.62% and

78.53%, respectively. The LSTM and 1D-CNN models also show consistent and significant improvements across most classes, although there is a slight decline for the Benign and Theft classes, respectively.

To summarise the performance variations observed in multi-class classification following the inclusion of temporal features, the results show that although the magnitude of improvement varies across individual classes, the overall macro-averaged improvement remains consistently positive across all datasets and models. This finding demonstrates that incorporating temporal features generally enhances classification performance. It is also worth noting that not all attack types exhibit strong temporal dependencies, and the results accurately reflect this characteristic.

TABLE 7. Feature groups used in the permutation analysis.

Group	Feature Name
Durations	FLOW_DURATION_MILLISECONDS
	DURATION_IN
	DURATION_OUT
Timestamps	FLOW_START_MILLISECONDS
	FLOW_END_MILLISECONDS
IAT	SRC_TO_DST_IAT_MIN
	SRC_TO_DST_IAT_MAX
	SRC_TO_DST_IAT_AVG
	SRC_TO_DST_IAT_STDDEV
	DST_TO_SRC_IAT_MIN
	DST_TO_SRC_IAT_MAX
	DST_TO_SRC_IAT_AVG
	DST_TO_SRC_IAT_STDDEV

C. PERMUTATION ANALYSIS

The initial comparative evaluations between models with and without temporal features are essential to establish the practical impact of incorporating these features across different architectures and datasets. This stage provides a broad understanding of how temporal information influences model performance under realistic conditions. However, such evaluations alone cannot fully isolate the specific contribution

of temporal features, as the model architectures and input representations inevitably change when new features are introduced. After confirming the overall effect in the previous section, we conduct a permutation analysis [26] in this section, as a complementary step to more accurately quantify the role of temporal features within a fixed model. This two-stage approach provides both empirical validation through direct performance comparison and causal insight through controlled feature importance analysis.

Permutation analysis is a model-agnostic technique that estimates the contribution of a feature (or group of features) by measuring the degradation in classification performance when the feature values are randomly shuffled. In our permutation analysis, we combine the related features into three groups as shown in Table 7.

The Durations group includes three features, that existed in the previous versions of the datasets. The Timestamps group includes the flow start and end times (in milliseconds), and the IAT group includes all inter-arrival time statistics for source-to-destination and destination-to-source flows.

We first shuffle each group independently to examine its individual contributions to classification performance. Next, we shuffle all temporal features simultaneously to evaluate their collective importance. These experiments are performed for both binary and multi-class classification tasks. If shuffling the temporal features results in a substantial drop in performance, it indicates that the models rely on temporal information; if it leads to little or no change, it suggests that the models do not strongly depend on these features.

Figure 7 shows the results of the permutation analysis for the binary classification task across the four NF3 datasets using the LSTM, 1D-CNN, and Transformer models. The solid orange bars represent the baseline F1 scores with unshuffled data, while the subsequent bars illustrate performance when timestamps, durations, IAT, and all temporal features are shuffled.

Across all datasets, shuffling temporal features leads to noticeable performance degradation, confirming that the models rely on temporal information to varying degrees. For NF-UNSW, all models achieve strong baselines (94–99% F1), but timestamp shuffling causes a sharp decline, most notably for the LSTM (47.19%), while 1D-CNN and Transformer remain more robust (83.63% and 98.66%). In NF-ToN, where baseline scores are 93–95%, timestamp shuffling again results in substantial drops (around 63.98%), whereas IAT and duration shuffling have smaller effects, suggesting timestamps are the dominant temporal factor. For NF-CIC, all temporal groups contribute significantly; LSTM performance falls from 96.66% to between 60–74%, with similar declines for CNN and Transformer (65.56% and 72.70%), showing that temporal dynamics are central to this dataset. In contrast, NF-BoT remains almost unaffected (around 99%), reflecting its severe class imbalance and limited temporal dependency.

Overall, the analysis demonstrates that temporal features are vital for effective binary attack detection in most

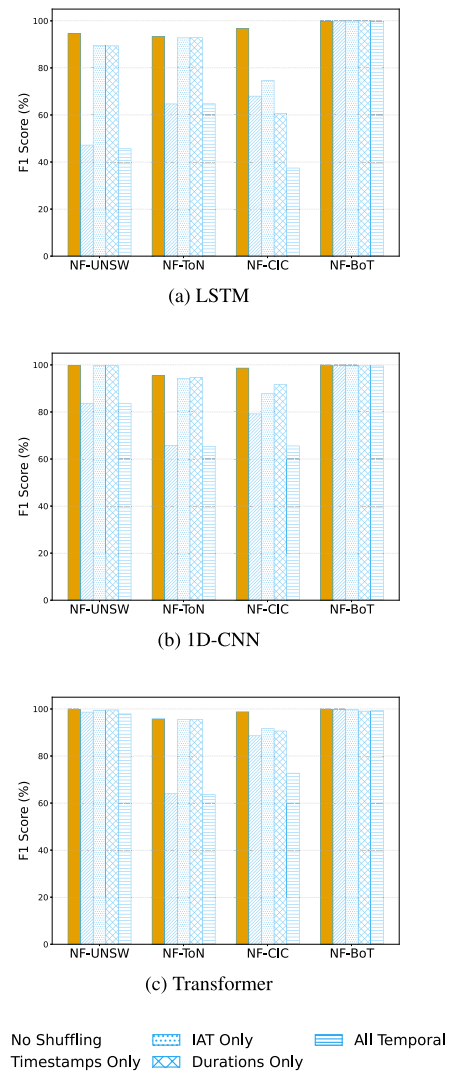


FIGURE 7. Permutation analysis for the binary classification task across the NF3 datasets reported using F1 score. The orange bars denote the baseline performance using unshuffled data with all features.

datasets. The 1D-CNN and LSTM models show higher sensitivity to temporal disruption, whereas the Transformer remains relatively stable, likely due to its ability to capture broader contextual dependencies even when temporal order is distorted.

Figure 8 presents the results of the permutation analysis for the multi-class classification task across the four NF3 datasets using the LSTM, 1D-CNN, and Transformer architectures. Results are reported using **macro F1 scores**, which better reflect the impact of class imbalance by averaging performance across all classes. The orange bars represent the baseline performance using all NF3 features, while the subsequent bars show the impact of shuffling timestamps, durations, IAT, and all temporal features.

Across all datasets, shuffling temporal features reduces detection performance to varying degrees, confirming their essential role in distinguishing attack categories. For NF-UNSW, baseline macro F1 scores are modest (39.95% for

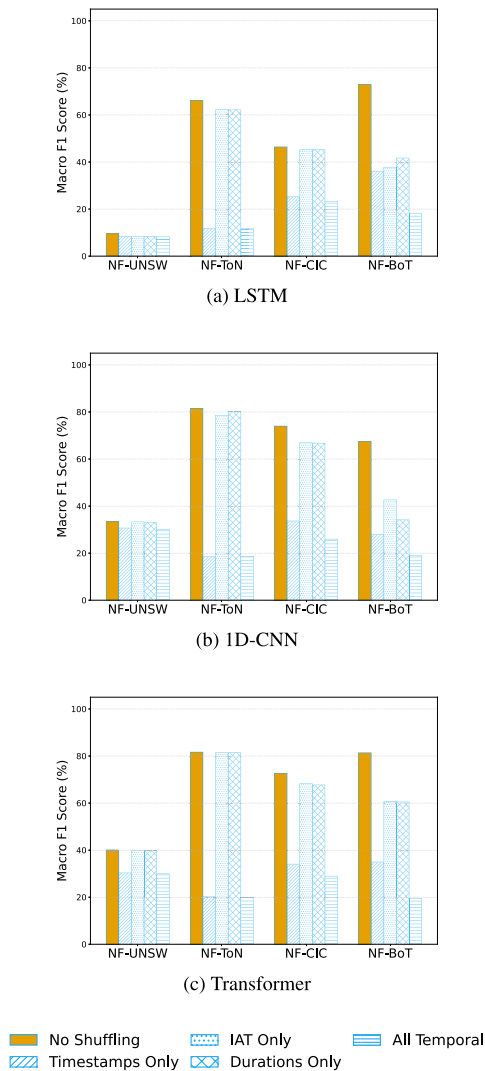


FIGURE 8. Permutation analysis for the multi-class classification task across the NF3 datasets reported using macro F1 score. The orange bars denote the baseline performance using unshuffled data with all features.

Transformer, 33.34% for 1D-CNN, and below 10% for LSTM), and shuffling any temporal group produces only small changes. A possible contributing factor to the weak performance and low sensitivity to temporal shuffling in NF-UNSW is the small sample size of several attack classes. Many attack classes in this dataset contain only a few hundred or even fewer examples, which makes it difficult for the models to learn stable temporal patterns.

In NF-ToN, where baselines are higher (around 66–81%), timestamp shuffling leads to a severe drop in all models, reducing the Transformer and 1D-CNN to approximately 18% and the LSTM to 11.72%, while shuffling IAT or duration features has a milder impact, keeping scores above 60%. For NF-CIC, removing timestamps cuts performance by nearly half or even less, with IAT and duration shuffling also causing notable declines (around 45.21% for LSTM and 66–67% for 1D-CNN and Transformer). When all temporal features are shuffled together, macro F1 falls below 30%, showing that temporal structure is vital for this dataset.

In NF-BoT, unlike the binary classification case where all models maintained near-perfect performance after shuffling, the effect becomes clearly visible in the multi-class setting. Here, timestamp permutation halves performance across all models, and shuffling all temporal features reduces scores to about 18–19%, revealing the underlying importance of temporal dependencies that was previously masked by the dataset's strong class imbalance.

Overall, these results demonstrate that temporal features are essential for effective multi-class attack detection. Among the three temporal groups, timestamps and IAT, which were newly introduced in the NF3 datasets, consistently emerge as the most influential components. In contrast, duration features, which were already included in earlier NF2 versions, have a smaller impact on performance. The macro F1 metric further highlights how dataset imbalance interacts with temporal dependencies, reinforcing the critical role of these newly added temporal features in shaping overall model performance.

VII. CONCLUSION

This study provides a systematic examination of the role of temporal information in flow-based network intrusion detection. By extending existing NetFlow benchmarks to create the NF3 datasets, two new temporal feature groups, timestamps and IAT statistics, were added to the previously available duration features. This enhancement enabled a detailed investigation of how time-dependent patterns influence both binary and multi-class attack detection. Comparative evaluations and permutation analyses conducted across four benchmark datasets showed that temporal features capture behavioural patterns that time-agnostic flow features alone cannot represent. Among these, timestamps and IAT features consistently proved to be the most informative, allowing finer distinction between benign and malicious traffic. In contrast, duration features, which were already included in NF2, contributed less to overall performance improvements.

Experiments using sequential deep learning models confirmed that including temporal features improves detection accuracy across different architectures and datasets. The improvement was particularly notable in multi-class classification, where temporal information helps models separate multiple attack types more effectively.

This study demonstrates that temporal features substantially enhance flow-based intrusion detection. Using NF3 datasets, we show that timestamps and IAT metrics yield meaningful performance improvements across LSTM, 1D-CNN, and Transformer architectures—particularly for attacks with strong temporal characteristics. The permutation analysis confirmed their causal importance. Duration features alone were less impactful, highlighting the added value of fine-grained time features. NF3 establishes a unified, time-aware benchmark for future research in sequence modelling, temporal embeddings, and cross-dataset generalisation for real-world NIDS deployment.

REFERENCES

- [1] N. Hoque, M. H. Bhuyan, R. C. Baishya, D. K. Bhattacharyya, and J. K. Kalita, "Network attacks: Taxonomy, tools and systems," *J. Netw. Comput. Appl.*, vol. 40, pp. 307–324, Apr. 2014.
- [2] O. Yoachimik and J. Pacheco. (Apr. 2025). *Cloudflare's 2025 Q1 Ddos Threat Report*. [Online]. Available: <https://blog.cloudflare.com/ddos-threat-report-for-2025-q1/>
- [3] G. Vormayr, J. Fabini, and T. Zseby, "Why are my flows different? A tutorial on flow exporters," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 2064–2103, 3rd Quart., 2020.
- [4] B. Claise, "Cisco systems netflow services export version 9," Cisco Syst., San Jose, CA, USA, Tech. Rep. 838, 2004.
- [5] M. F. Umer, M. Sher, and Y. Bi, "Flow-based intrusion detection: Techniques and challenges," *Comput. Secur.*, vol. 70, pp. 238–254, Sep. 2017.
- [6] D. Zhou, Z. Yan, Y. Fu, and Z. Yao, "A survey on network data collection," *J. Netw. Comput. Appl.*, vol. 116, pp. 9–23, Aug. 2018.
- [7] S. Kumar, S. Gupta, and S. Arora, "Research trends in network-based intrusion detection systems: A review," *IEEE Access*, vol. 9, pp. 157761–157779, 2021.
- [8] M. Roesch, "Snort: Lightweight intrusion detection for networks," in *Proc. Lisa*, vol. 99, 1999, pp. 229–238.
- [9] R. A. Disha and S. Waheed, "Performance analysis of machine learning models for intrusion detection system using Gini impurity-based weighted random forest (GIWRF) feature selection technique," *Cybersecurity*, vol. 5, no. 1, p. 1, Dec. 2022.
- [10] S. Hosseiniinorbini, S. Layeghy, M. Sarhan, R. Jurdak, and M. Portmann, "Exploring edge TPU for network intrusion detection in IoT," *J. Parallel Distrib. Comput.*, vol. 179, Sep. 2023, Art. no. 104712.
- [11] G. Apruzzese, L. Pajola, and M. Conti, "The cross-evaluation of machine learning-based network intrusion detection systems," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 4, pp. 5152–5169, Dec. 2022.
- [12] S. Omar, A. Ngadi, and H. H. Jebur, "Machine learning techniques for anomaly detection: An overview," *Int. J. Comput. Appl.*, vol. 79, no. 2, pp. 33–41, Oct. 2013.
- [13] S. Duque Anton, L. Ahrens, D. Fraunholz, and H. D. Schotten, "Time is of the essence: Machine learning-based intrusion detection in industrial time series data," in *Proc. IEEE Int. Conf. Data Mining Workshops (ICDMW)*, Nov. 2018, pp. 1–6.
- [14] A. Corsini, S. J. Yang, and G. Apruzzese, "On the evaluation of sequential machine learning for network intrusion detection," in *Proc. 16th Int. Conf. Availability, Rel. Secur.*, New York, NY, USA, Aug. 2021, pp. 1–10.
- [15] M. A. Hogo, "Temporal analysis of intrusion detection," in *Proc. Int. Carnahan Conf. Secur. Technol. (ICCST)*, Oct. 2014, pp. 1–6.
- [16] N. Moustafa and J. Slay, "The significant features of the UNSW-NB15 and the KDD99 data sets for network intrusion detection systems," in *Proc. 4th Int. Workshop Building Anal. Datasets Gathering Exper. Returns Secur. (BADGERS)*, Nov. 2015, pp. 25–31.
- [17] M. Awad and S. Fraihat, "Recursive feature elimination with cross-validation with decision tree: Feature selection method for machine learning-based intrusion detection systems," *J. Sensor Actuator Netw.*, vol. 12, no. 5, p. 67, Sep. 2023.
- [18] A. Ayantayo, A. Kaur, A. Kour, X. Schmoor, F. Shah, I. Vickers, P. Kearney, and M. M. Abdelsamea, "Network intrusion detection using feature fusion with deep learning," *J. Big Data*, vol. 10, no. 1, p. 167, Nov. 2023.
- [19] L. D. Manocchio, S. Layeghy, W. W. Lo, G. K. Kulatilleke, M. Sarhan, and M. Portmann, "FlowTransformer: A transformer framework for flow-based network intrusion detection systems," *Expert Syst. Appl.*, vol. 241, May 2024, Art. no. 122564.
- [20] M. Sarhan, S. Layeghy, N. Moustafa, M. Portmann, "NetFlow datasets for machine learning-based network intrusion detection systems," *International Conference on Big Data Technologies and Applications*. Cham, Switzerland: Springer, 2021, pp. 117–135.
- [21] M. Sarhan, S. Layeghy, and M. Portmann, "Towards a standard feature set for network intrusion detection system datasets," *Mobile Netw. Appl.*, vol. 27, no. 1, pp. 357–370, Feb. 2022.
- [22] (2025). *NetFlow V3 NIDS Datasets—The University of Queensland*. [Online]. Available: <https://staff.itee.uq.edu.au/marius/NIDSdatasets/>
- [23] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [24] E. U. H. Qazi, A. Almorjan, and T. Zia, "A one-dimensional convolutional neural network (1D-CNN) based deep learning system for network intrusion detection," *Appl. Sci.*, vol. 12, no. 16, p. 7986, Aug. 2022.
- [25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [26] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [27] A. Binbusayyis and T. Vaiyapuri, "Comprehensive analysis and recommendation of feature evaluation measures for intrusion detection," *Heliyon*, vol. 6, no. 7, Jul. 2020, Art. no. e04262.
- [28] M. Sarhan, S. Layeghy, and M. Portmann, "Evaluating standard feature sets towards increased generalisability and explainability of ML-based network intrusion detection," *Big Data Res.*, vol. 30, Nov. 2022, Art. no. 100359.
- [29] D. Choi, J. Rhee, and H. Park, "Attack-specific feature analysis framework for NetFlow IoT datasets," *Comput. Secur.*, vol. 157, Oct. 2025, Art. no. 104536.
- [30] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Comput. Intell. Secur. Defense Appl.*, Jul. 2009, pp. 1–6.
- [31] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6.
- [32] S. Lundberg and S. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4765–4774.
- [33] A. Iorliam, S. Tirunagari, A. T. S. Ho, S. Li, A. Waller, and N. Poh, "Flow size difference' can make a difference: Detecting malicious TCP network flows based on Benford's law," 2016, *arXiv:1609.04214*.
- [34] E. Biglar Beigi, H. Hadian Jazi, N. Stakhanova, and A. A. Ghorbani, "Towards effective feature selection in machine learning-based botnet detection approaches," in *Proc. IEEE Conf. Commun. Netw. Secur.*, Oct. 2014, pp. 247–255.
- [35] L. M. Chen, S.-W. Hsiao, M. C. Chen, and W. Liao, "Slow-paced persistent network attacks analysis and detection using spectrum analysis," *IEEE Syst. J.*, vol. 10, no. 4, pp. 1326–1337, Dec. 2016.
- [36] J. Zimmermann, A. J. Clark, G. Mohay, F. Pouget, and M. Daciér, "The use of packet inter-arrival times for investigating unsolicited internet traffic," in *Proc. 1st Int. Workshop Systematic Approaches Digit. Forensic Eng.*, 2006, pp. 89–104.
- [37] K. C. Gross, K. Vaidyanathan, and A. E. Brownsword, "Inter-arrival time intrusion-detection technique to provide enhanced cybersecurity, July 3 2018," U.S. Patent 10015 139, Jul. 3, 2018.
- [38] T. Benson, A. Akella, and D. A. Maltz, "Network traffic characteristics of data centers in the wild," in *Proc. 10th ACM SIGCOMM Conf. Internet Meas.*, Nov. 2010, pp. 267–280.
- [39] S. Kandula, S. Sengupta, A. Greenberg, P. Patel, and R. Chaiken, "The nature of data center traffic: Measurements & analysis," in *Proc. 9th ACM SIGCOMM Conf. Internet Meas.*, New York, NY, USA, Nov. 2009, pp. 202–208.
- [40] A. Lakhina, K. Papagiannaki, M. Crovella, C. Diot, E. D. Kolaczky, and N. Taft, "Structural analysis of network traffic flows," in *Proc. joint Int. Conf. Meas. Model. Comput. Syst.*, New York, NY, USA, Jun. 2004, pp. 61–72.
- [41] B. Li, J. Springer, G. Bebis, and M. Hadi Gunes, "A survey of network flow applications," *J. Netw. Comput. Appl.*, vol. 36, no. 2, pp. 567–581, Mar. 2013.
- [42] X. Han, R. Yin, Z. Lu, B. Jiang, Y. Liu, S. Liu, C. Wang, and N. Li, "STIDM: A spatial and temporal aware intrusion detection model," in *Proc. IEEE 19th Int. Conf. Trust, Secur. Privacy Comput. Commun. (TrustCom)*, Dec. 2020, pp. 370–377.
- [43] Y. Zhang, X. Chen, L. Jin, X. Wang, and D. Guo, "Network intrusion detection: Based on deep hierarchical network and original flow data," *IEEE Access*, vol. 7, pp. 37004–37016, 2019.
- [44] J. Zhao, R. Masood, and S. Seneviratne, "A review of computer vision methods in network security," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1838–1878, 3rd Quart., 2021.
- [45] C. Wu, J. Sun, J. Chen, M. Alazab, Y. Liu, and Y. Xiang, "TCG-IDS : Robust network intrusion detection via temporal contrastive graph learning," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 1475–1486, 2025.
- [46] Y. Wang, J. Li, W. Zhao, Z. Han, H. Zhao, L. Wang, and X. He, "N-STGAT: Spatio-temporal graph neural network based network intrusion detection for near-Earth remote sensing," *Remote Sens.*, vol. 15, no. 14, p. 3611, Jul. 2023.

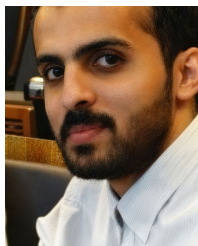
- [47] M. Hamdan, B. Mohammed, U. Humayun, A. Abdelaziz, S. Khan, M. A. Ali, M. Imran, and M. N. Marsono, "Flow-aware elephant flow detection for software-defined networks," *IEEE Access*, vol. 8, pp. 72585–72597, 2020.
- [48] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," *ICISSp*, vol. 1, pp. 108–116, Jan. 2018.
- [49] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets," *Sustain. Cities Soc.*, vol. 72, Sep. 2021, Art. no. 102994.
- [50] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Gener. Comput. Syst.*, vol. 100, pp. 779–796, Nov. 2019.
- [51] *NProbe, an Extensible Netflow V5/V9/IPFIX Probe for IPV4/V6*, Ntop, New York, NY, USA, 2017.
- [52] M. Luay, S. Layeghy, Y. Pandey, G. Kulatilleke, and M. Portmann, "Multimodal LLMs for zero-shot intrusion detection using NetFlow visualisations," in *Proc. IEEE 50th Conf. Local Comput. Netw. (LCN)*, Oct. 2025, pp. 1–7.
- [53] L. Manocchio, S. Layeghy, M. Gallagher, and M. Portmann. (2023). *An Evaluation of Preprocessing Methods for Machine Learning Based NIDS*. [Online]. Available: <https://ssrn.com/abstract=4647372>
- [54] M. Sarhan, S. Layeghy, N. Moustafa, M. Gallagher, and M. Portmann, "Feature extraction for machine learning-based intrusion detection in IoT networks," *Digit. Commun. Netw.*, vol. 10, no. 1, pp. 205–216, Feb. 2024.
- [55] X. Wang, J. Liu, and C. Zhang, "Network intrusion detection based on multi-domain data and ensemble-bidirectional LSTM," *EURASIP J. Inf. Secur.*, vol. 2023, no. 1, p. 5, Jun. 2023.



MOHANAD SARHAN received the bachelor's degree in information technology and the Ph.D. degree in AI security from The University of Queensland. He has authored and co-authored peer-reviewed publications in the areas of information security and applied AI, contributing to both academic scholarship and industry-relevant knowledge. His research focuses on cybersecurity, artificial intelligence security, and the protection of digital systems.



GAYAN KULATILLEKE received the Bachelor of Engineering degree in computer science and engineering from the University of Moratuwa, Sri Lanka, the master's degree in big data and AI from the Queen Mary University of London, U.K., and the Ph.D. degree from the School of Information Technology and Electrical Engineering (ITEE), The University of Queensland, Australia. His research interests include graph neural networks, algorithm optimization, LLMs, network security, and AI/ML applications in cybersecurity.



application of machine learning and large language models to cybersecurity.

MAJED LUAY received the bachelor's degree from Taif University, Saudi Arabia, in 2015, and the Master of Information Security degree from the Queensland University of Technology (QUT), Brisbane, in 2020. He is currently pursuing the Ph.D. degree with The University of Queensland (UQ), Australia. His research interests include network security, flow-based intrusion detection, Denial-of-Service (DoS/DDoS) attack analysis, visual analytics for network traffic, and the



cybersecurity, and the Internet of Things.

SIAMAK LAYEGHY received the Ph.D. degree in computer science from The University of Queensland, Australia, where his research focused on constraint programming approaches for software-defined networking. He is a Lecturer and a Researcher in applied machine learning and network security at the School of Electrical Engineering and Computer Science, The University of Queensland. His research interests include applied machine learning, computer networks,



world applications. Her research interests include deep learning, edge AI, and the Internet of Things (IoT), with an emphasis on scalable, and robust AI systems for critical infrastructure.

NILOUFAR NOORBIN received the bachelor's degree in electronics engineering and the master's degree in control engineering, building a strong foundation in systems and automation, and the Ph.D. degree in computer science from The University of Queensland, focusing on deep learning at the edge. She is currently a Principal Machine Learning professional at Sydney Water, where she leads the design and deployment of machine learning solutions for large-scale, real-



MARIUS PORTMANN (Member, IEEE) received the Ph.D. degree in technical sciences (electrical engineering) from the Swiss Federal Institute of Technology (ETH Zürich). He is a Professor and the Cisco Chair in network security at the School of Electrical Engineering and Computer Science, The University of Queensland, Australia. His research interests include computer networks, cybersecurity, the Internet of Things, and applied AI.

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